The Meßthaler-Wulff Project

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Blazingly fast code for finding all crystals (subsets of a graph) that can be constructed using only transformations that locally minimize surface energy.

The Problem

Let (G, E) be some graph and $\mathcal{N} : G \to \wp(G)$ denote the neighbors of a given node, defined as

$$\mathcal{N}(n) \coloneqq \{n_0 \in G \mid \{n_0, n\} \in E\}$$

Now we can define a crystal as $c \subset G$ with c finite. This allows us to define the set of all crystals $C \coloneqq \{c \subset G \mid c \text{ finite}\}$.

Now we can define the surface energy of the crystal c (or arbitrary subsets of G) as

$$E_c \coloneqq \sum_{n \in c} f_{G \setminus c}(n)$$

where $f_M(n)$ denotes the "friendliness" of the node n or how many friends it has defined as

$$f_M(n) \coloneqq \#\{n_0 \in \mathcal{N}(n) \mid n_0 \in M\}$$

The idea now is to find crystals c such that $\frac{E_c}{\#c}$ is optimal.

`Note:`

$$E_c = \sum_n \chi_c(n) \cdot (\#\mathcal{N}(n) - f_c(n))$$

$$= \sum_n \chi_c(n) \cdot \#\mathcal{N}(n) - \sum_{a,b \in \mathcal{N}(a)} \chi_c(a) \cdot \chi_c(b)$$

$$\mathcal{N}' \coloneqq (\#\mathcal{N}(n_i))_{1 \leq i \leq B}$$

$$E_c = (\mathcal{N}'^\top \cdot \chi_c) - (\chi_c^\top \cdot A_G \cdot \chi_c)$$

The Crystal Graph

We can impose a graph structure on C with the edges

$$T \coloneqq \{\{c, c \setminus \{n\}\} \mid c \in C \text{ and } n \in c\}$$

we call these the transformations and (C, T) the transformation graph.

If we want to discover optimal crystals, then we must efficiently walk this graph. Since this graph is very dense and very large, we must first discuss some optimizations:

- $O(1)$ transformations

Efficient Transformations

In the code this is achieved using the stateful class `AdditiveSimulation` that walks along the transformation graph. This class keeps track of two important properties, namely $f_c(n)$ and $\chi_c(n)$ for (almost) all $n \in G$, we do not need to store the (possibly) infinite number of values, as we can let $f_c(n)$ default to 0 and $\chi_c(n)$ to 0. $\chi_c(n) \in \{0, 1\}$ is the characteristic function of the set c and is equal to 1 exactly iff the node is in c.

To the characteristic function we can associate a sign using $s : \{0, 1\} \to \{-1, 1\}$, defined as $s(x) = 2x - 1$.

Now let us take a look at what happens, if we walk along an edge $t \in T$. Let n be the single node that is affected by this transformation.

Let φ be any specific integer quantity we are tracking, then let $\Delta\varphi \coloneqq \varphi' - \varphi$ be the difference between the old and the new value of the quantity.

χ_c is updated using $\Delta\chi_c(n) = -s \circ \chi_c(n)$.

The friendliness $f_c(n)$ does not change. Instead, the friendliness of each neighbor must be updated. So for each $n_0 \in \mathcal{N}(n)$: $\Delta f_c(n_0) = -s \circ \chi_c(n)$.

For updating the Energy E_c will first require some work. We already know the update rule for neighbors n_0 of n. For nodes n_0 that are not n or neighbors of n the differential is $\Delta f_c(n_0) = 0$, in essence this is also not problematic.

However for n the differential is more involved, as the transition makes the node become part of or be excluded from the sum that defines E_c. So its contribution is

$$-s \circ \chi_c(n) \cdot f_{G \setminus c}(n) = -s \circ \chi_c(n) \cdot [\#\mathcal{N}(n) - f_c(n)]$$

Now for the calculation:

$$\Delta E_c = \sum_{n_0 \in c} f_{G \setminus c}(n_0)$$

$$= \left[\sum_{n_0 \in \mathcal{N}(n) \cap c} \Delta f_{G \setminus c}(n_0) \right] - s \circ \chi_c(n) \cdot [\#\mathcal{N}(n) - f_c(n)]$$

$$= \left[\sum_{n_0 \in \mathcal{N}(n) \cap c} s \circ \chi_c(n) \right] - s \circ \chi_c(n) \cdot [\#\mathcal{N}(n) - f_c(n)]$$

$$= s \circ \chi_c(n) \cdot f_c(n) - s \circ \chi_c(n) \cdot [\#\mathcal{N}(n) - f_c(n)]$$

$$= s \circ \chi_c(n) \cdot (2f_c(n) - \#\mathcal{N}(n))$$

Min-Calculus

$$Q : \mathbb{Z}^n \to \mathbb{Z}$$

$$r_k : \mathbb{Z}^{n-k} \to \mathbb{Z}$$

$$\min(Q) = r_n$$

$$r_0 = Q$$

$$r_k = \min(r_{k-1}(0), r_{k-1}(1))$$

Solvers